Predicted vs measured epistasis of double mutants (seed 1; ρ in each panel; dashed = y = x)

ESM-MSR (baseline) epistasis head (MT − WT)

ESM-MSR (baseline) combined, vs additive

MutateEverything st2 (vs st1 singles)

ThermoMPNN-D (vs ThermoMPNN)

SPURS multi (vs SPURS singles)

Tsuboyama test measured dddG (kcal/mol)

GB1 measured dddG (kcal/mol)

GRB2 abund. measured dddG (kcal/mol)

predicted epistasis (kcal/mol)

	ESM-MSR (baseline) epistasis head (MT − WT)	ESM-MSR (baseline) combined, vs additive	MutateEverything st2 (vs st1 singles)	ThermoMPNN-D (vs ThermoMPNN)	SPURS multi (vs SPURS singles)
Tsuboyama test	ρ = 0.44 r = 0.42 n = 12,455	ρ = 0.35 r = 0.35 n = 12,455	ρ = 0.41 r = 0.45 n = 12,455	ρ = 0.10 r = 0.10 n = 12,455	ρ = 0.37 r = 0.36 n = 12,455
GB1	ρ = 0.51 r = 0.44 n = 1,939	ρ = 0.56 r = 0.53 n = 1,939	ρ = 0.36 r = 0.33 n = 1,939	ρ = 0.33 r = 0.22 n = 1,939	ρ = 0.47 r = 0.33 n = 1,939
GRB2 abund.	ρ = 0.34 r = 0.48 n = 62,189	ρ = 0.45 r = 0.57 n = 62,189	ρ = 0.35 r = 0.45 n = 62,189	ρ = 0.19 r = 0.26 n = 62,189	ρ = 0.39 r = 0.47 n = 62,189